Gaotian Wang

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Education

Sep 2022 - Present | **Rice University**, *Houston, TX*
Ph.D. in Computer Science
Advisor: Dr. Kaiyu Hang

Sep 2018 - Jun 2022 | **University of Science and Technology of China**, *Hefei, China*
B.S. in Optical Engineering and Computer Science
Advisor: Dr. Nikolaos M. Freris

Work Experience

Apr 2025 - Present | **XLeRobot** *Houston, TX*
Founder and Core Developer
◇ *An opensource dual-arm mobile home robot for **$660 cost***
◇ ***4k+ GitHub stars** and wide media coverage*
◇ *Collaboration with Huggingface, SEEED, Lightwheel AI, etc.*
Skills: *CAD modeling and 3D fabrication, Isaac Sim, Maniskill, embedded systems (Jetson Orin, Thor), inverse kinematics, VR interface, computer vision, etc.*

Research Interests

My research focuses on **robust non-prehensile manipulation** under uncertainty, **physics-aware multimodal world models**, and their integration for real-world applications, especially **low-cost home robots**.

Research Experience

Sep 2022 - Present | **RobotΠ Lab** *at Rice University, Houston, TX*
Graduate Student, Advisor: Prof. Kaiyu Hang
Topic: Robust nonprehensile manipulation with intuitive physics world models

◇ Developed ***Caging in Time*** and ***Caging in Dreams*** framework achieving zero-shot sim-to-real transfer for robust manipulation under perception uncertainties, with 100% success rate in open-loop **object pushing**, **dynamics ball balancing and catching**, and **deformable chain manipulation**.
◇ Designed diffusion-based **Task-Specific Intuitive Physics (TSIP) models** with in-representation domain randomization, achieving **58× computational speedup over GPU-accelerated simulators**.
◇ *Caging in Time* is published on **IJRR**, *Caging in Dreams* is in submission.
Skills: Maniskill, optimization (MPPI), planning, diffusion models.

Nov 2021 - Jun 2022 | **AIoT Lab** *at USTC, Hefei, China*
Undergraduate Researcher, Advisor: Prof. Nikolaos M. Freris
Topic: Kinodynamic planning and control for soft robotic arms
Skills: Mujoco, planning (RRT), mechanical modeling, control theories.

Apr 2021 - Sep 2021 | **Reconfigurable Robotics Lab** *at EPFL, Lausanne, Switzerland*
Guest Researcher, Supervisor: Prof. Jamie Paik
Topic: Origami structures modeling, simulation and hysteresis evaluation
Skills: Taichi language, material properties analysis, 3D fabrication.
The structure was deployed on Mercedes-Benz VISION AVTR.

Aug 2020 - Oct 2021 | **USTC Soft Robotics Lab** *at USTC, Hefei, China*
Undergraduate Researcher, Advisor: Prof. Xiaoping Chen
Topic: Soft robotic arms modeling and learning-based control
Skills: COMSOL, inverse kinematics, reinforcement learning.

Publications

† Equal Contribution.

In-submission

I1. Caging in Dreams: Zero-shot Object Manipulation via Diffusion-based Task-Specific Intuitive Physics and In-Representation Domain Randomization
Gaotian Wang†, Kejia Ren, Andrew S. Morgan, and Kaiyu Hang.

Peer-Reviewed Journal Papers

J4. Caging in Time: A Framework for Robust Object Manipulation under Large Uncertainties
Gaotian Wang†, Kejia Ren†, Andrew S. Morgan, and Kaiyu Hang.
The International Journal of Robotics Research (IJRR), 2025.

J3. Object-Centric Kinodynamic Planning for Nonprehensile Robot Rearrangement Manipulation
Kejia Ren, **Gaotian Wang**, Andrew S. Morgan, Lydia E. Kavraki, and Kaiyu Hang.
The IEEE Transactions on Robotics (T-RO), 2025.

J2. Collision-inclusive Manipulation Planning for Occluded Object Grasping via Compliant Robot Motions
Kejia Ren, **Gaotian Wang**, Andrew S. Morgan, and Kaiyu Hang.
The IEEE Robotics and Automation Letters (RAL), 2025.

J1. A Reinforcement Learning Method for Motion Control With Constraints on an HPN Arm
Yinghao Gan, Peijin Li, Hao Jiang, **Gaotian Wang**, Yusong Jin, Xiaoping Chen, and Jianmin Ji.
IEEE Robotics and Automation Letters (RAL), 2022

Peer-Reviewed Conference Papers

C6. rt-RISeg: Real-Time Model-Free Robot Interactive Segmentation for Active Instance-Level Object Understanding
Howard Qian, Yiting Chen, **Gaotian Wang**, Podshara Chanrungmaneekul, and Kaiyu Hang.
In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2025*

C5. UNO Push: Unified Nonprehensile Object Pushing via Non-Parametric Estimation and MPC
Gaotian Wang, Kejia Ren, and Kaiyu Hang.
In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024*

C4. RISeg: Robot Interactive Object Segmentation via Body Frame-Invariant Features
Howard Qian, Yangxiao Lu, Kejia Ren, **Gaotian Wang**, Ninad Khargonkar, Yu Xiang, and Kaiyu Hang.
In *IEEE International Conference on Robotics and Automation (ICRA) 2024*

C3. Kinematic Modeling and Control of a Soft Robotic Arm with Non-constant Curvature Deformation
Zhanchi Wang, **Gaotian Wang**, Xiaoping Chen, and Nikolaos M Freris.
In *IEEE International Conference on Robotics and Automation (ICRA) 2024*

C2. Dynamic Modeling and Control of a Soft Robotic Arm Using a Piecewise Universal Joint Model
Zhanchi Wang, **Gaotian Wang**, Xiaoping Chen, and Nikolaos M. Freris.
In *IEEE International Conference on Robotics and Biomimetics (ROBIO) 2023*

C1. A Q-Learning Control Method for a Soft Robotic Arm Utilizing Training Data from a Rough Simulator
Peijin Li, **Gaotian Wang**, Hao Jiang, Yusong Jin, Yinghao Gan, Xiaoping Chen, and Jianmin Ji.
In *IEEE International Conference on Robotics and Biomimetics (ROBIO) 2021*

Theses

T1. A Randomized Kinodynamic Planner for Soft Robots based on Piecewise Universal Joint Model.
Gaotian Wang. Bachelor's thesis, USTC, Hefei, China, 2022

Academic Service

Reviewer:
- ◇ ICRA 2025, IROS 2025, RAL
- ◇ ICRA 2024, RAL
- ◇ IROS 2023

Teaching

Fall 2023	Teaching Assistant for **Algorithmic Robotics** *COMP/ELEC/MECH 450/550* *at Rice University*
Spring 2023	Teaching Assistant for **Deep Learning for Vision & Language** *COMP 646* *at Rice University*
Fall 2020-2022	In-lab Teaching Assistant for **College Physics-Comprehensive Experimentation** *at University of Science and Technology of China*

Skills

Programming:	Python, C/C++, Matlab, linux, Taichi
Frameworks:	ROS, ROS2, PyTorch, OpenCV, OMPL, RViz, LeRobot
Simulators:	Maniskill, Isaac Sim, PyBullet, MuJoCo, Genesis
Robotics:	SE(3) transforms, optimization, control theory, computer vision, RL
Others:	CAD modeling and 3D fabrication, Blender

Scholarships and Certificates

2022	Mengzhilan (Dream Of Blue) China Aerospace Foundation Scholarship, USTC
2020	Outstanding Student Scholarship, *Gold* (Top 3%), USTC
2019	Endeavor Student Scholarship, USTC
2018	Yan Jici Talent Program Scholarship (Top 10%), USTC